

UNIVERSIDAD NACIONAL DE CÓRDOBA
FACULTAD DE CIENCIAS EXACTAS, FÍSICAS Y NATURALES

TEORÍA DE SEÑALES Y SISTEMAS LINEALES

PRÁCTICO: Problemas Capitulo 3- Continuación

ALEJANDRA LUCÍA PÉREZ LUCERO
GUSTAVO LAZARTE

SERIE DE FOURIER PARA SEÑALES DISCRETAS

BIBLIOGRAFÍA

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3.27. Una señal periódica discreta $x[n]$ es de valor real y tiene periodo fundamental $N=5$. Los coeficientes de la serie de Fourier diferente de cero para $x[n]$ son:

$$a_0 = 2, \quad a_2 = a_{-2}^* = 2 e^{j\frac{\pi}{6}}, \quad a_4 = a_{-4}^* = e^{j\frac{\pi}{3}}$$

Expresa $x[n]$ en la forma:

$$x[n] = A_0 + \sum_{k=1}^{\infty} A_k \sin(\omega_k n + \phi_k)$$

$$a_0 = 2$$

$$a_2 = a_{-2}^* = 2 e^{j\frac{\pi}{6}}$$

$$a_4 = a_{-4}^* = e^{j\frac{\pi}{3}}$$

$$a_2 = 2 e^{j\frac{\pi}{6}}$$

$$a_4 = e^{j\frac{\pi}{3}}$$

$$a_{-2} = 2 e^{-j\frac{\pi}{6}}$$

$$a_{-4} = 2 e^{-j\frac{\pi}{3}}$$

$$x[n] \text{ real ; } N = 5$$

Solución:

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{N}n}$$

$$x[n] = a_0 + a_2 e^{j2\left(\frac{2\pi}{5}\right)n} + a_{-2} e^{-j2\left(\frac{2\pi}{5}\right)n} + a_4 e^{j4\left(\frac{2\pi}{5}\right)n} + a_{-4} e^{-j4\left(\frac{2\pi}{5}\right)n}$$

Reemplazando, resulta:

$$x[n] = 2 + 2e^{j\frac{\pi}{6}} e^{j\frac{4\pi}{5}n} + 2e^{-j\frac{\pi}{6}} e^{-j\frac{4\pi}{5}n} + e^{j\frac{\pi}{3}} e^{j\frac{8\pi}{5}n} + e^{-j\frac{\pi}{3}} e^{-j\frac{8\pi}{5}n}$$

Agrupando exponenciales, resulta:

$$x[n] = 2 + 2e^{j\left(\frac{\pi}{6} + \frac{4\pi}{5}n\right)} + 2e^{-j\left(\frac{\pi}{6} + \frac{4\pi}{5}n\right)} + e^{j\left(\frac{\pi}{3} + \frac{8\pi}{5}n\right)} + e^{-j\left(\frac{\pi}{3} + \frac{8\pi}{5}n\right)}$$

$$x[n] = 2 + 2 \cdot \underset{2}{2} \left[\frac{e^{j\left(\frac{\pi}{6} + \frac{4\pi}{5}n\right)} + e^{-j\left(\frac{\pi}{6} + \frac{4\pi}{5}n\right)}}{2} \right] + \underset{2}{2} \left[\frac{e^{j\left(\frac{\pi}{3} + \frac{8\pi}{5}n\right)} + e^{-j\left(\frac{\pi}{3} + \frac{8\pi}{5}n\right)}}{2} \right]$$

Aplicando Euler, resulta:

$$x[n] = 2 + 4 \cos \left[\left(\frac{\pi}{6} + \frac{4\pi}{5}n \right) \right] + 2 \cos \left[\left(\frac{\pi}{3} + \frac{8\pi}{5}n \right) \right]$$

Calculo Auxiliar

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

y

$$\cos \theta = \sin(90 + \theta)$$

$$\frac{\pi}{6} + \frac{\pi}{2} = \frac{\pi + 3\pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3}$$

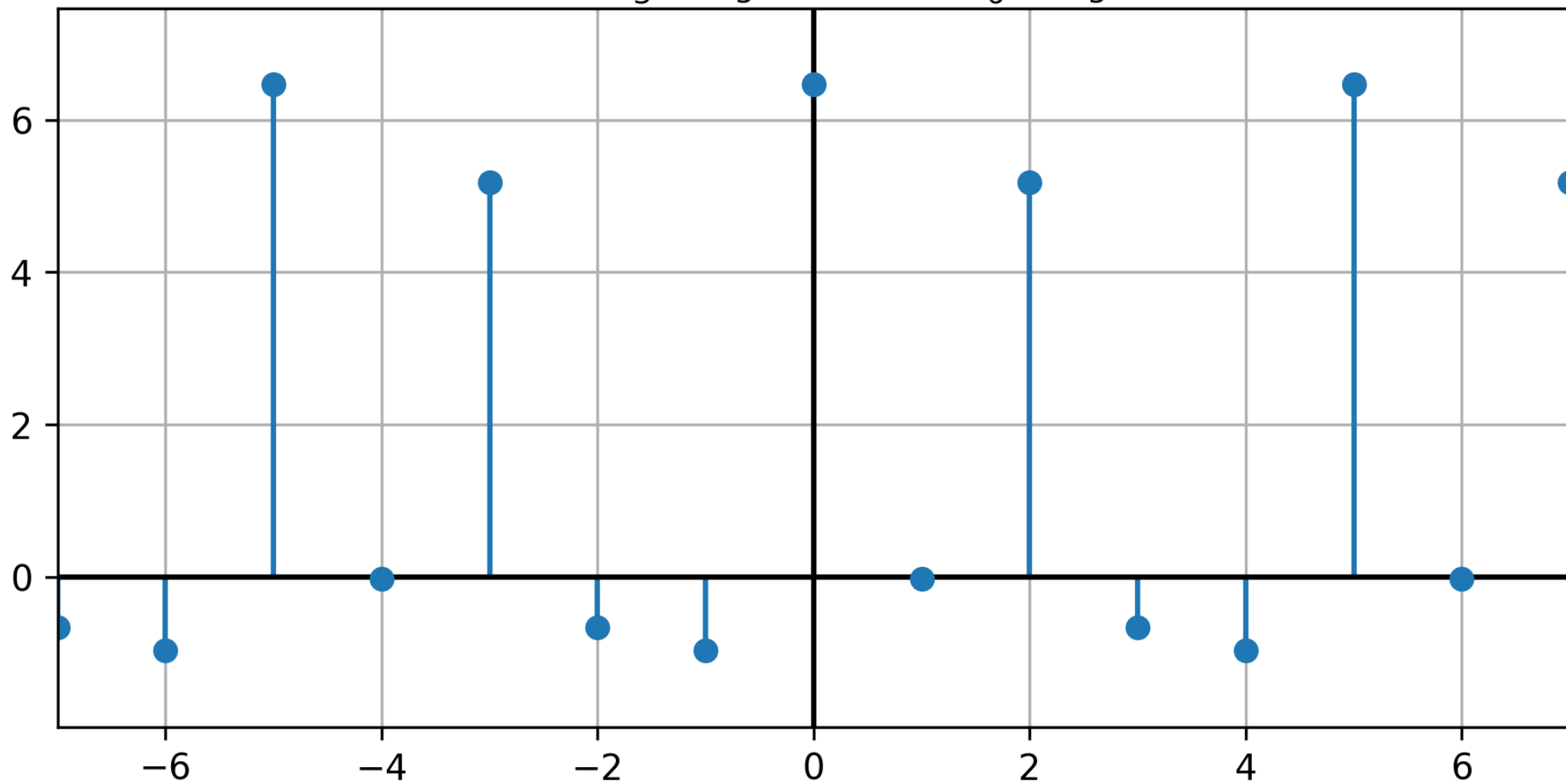
$$\frac{\pi}{3} + \frac{\pi}{2} = \frac{2\pi + 3\pi}{6} = \frac{5\pi}{6}$$

$$\cos \left[\frac{\pi}{6} + \frac{4\pi}{5} n \right] = \sin \left[\frac{2\pi}{3} + \frac{4\pi}{5} n \right]$$

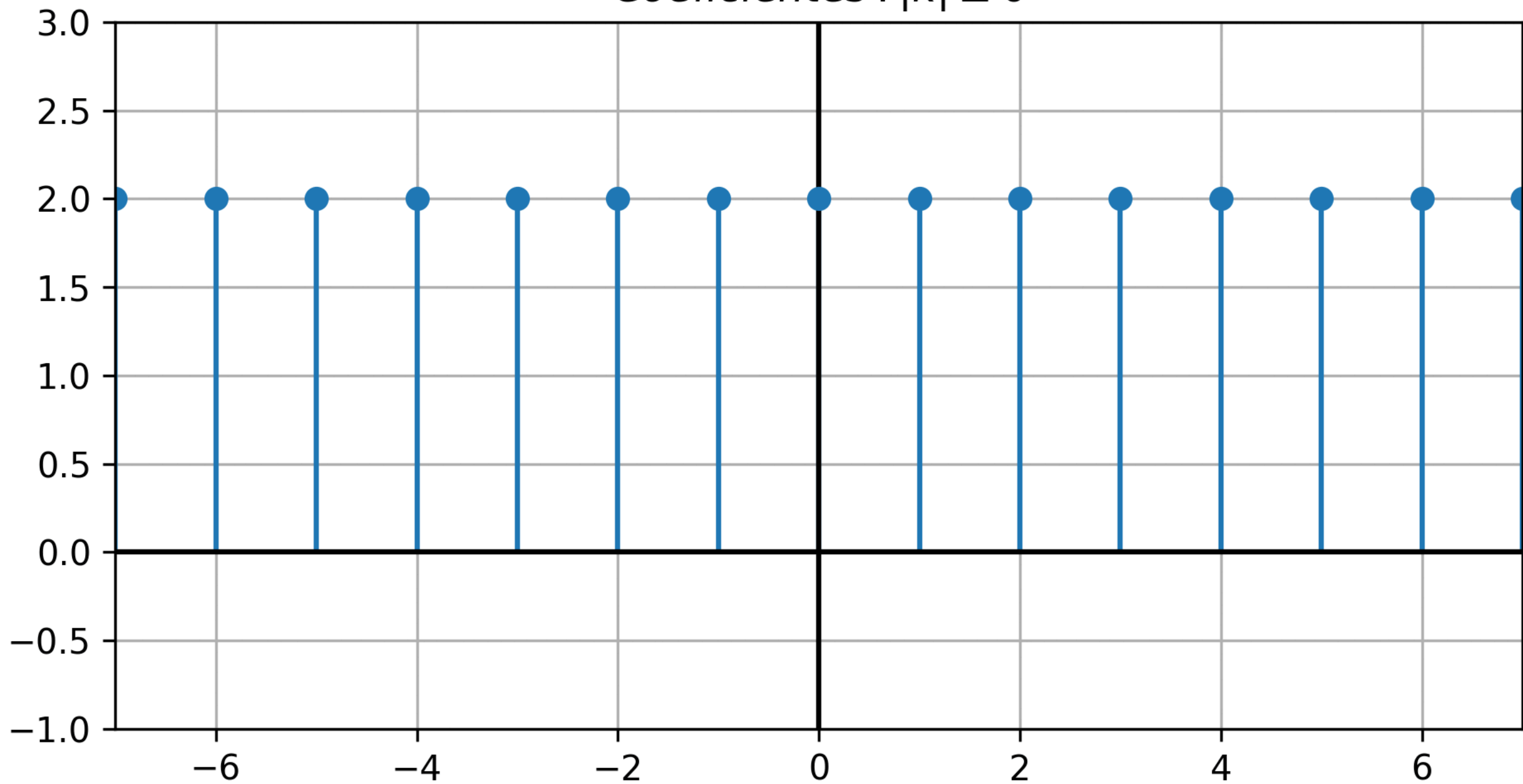
$$\cos \left[\frac{\pi}{3} + \frac{8\pi}{5} n \right] = \sin \left[\frac{5\pi}{6} + \frac{8\pi}{5} n \right]$$

$$x[n] = 2 + 4 \sin \left[\frac{2\pi}{3} + \frac{4\pi}{5} n \right] + 2 \sin \left[\frac{5\pi}{6} + \frac{8\pi}{5} n \right]$$

$$2 + 4\sin\left(\frac{2}{3}\pi + \frac{4}{5}\pi n\right) + 2\sin\left(\frac{5}{6}\pi + \frac{8}{5}\pi n\right)$$

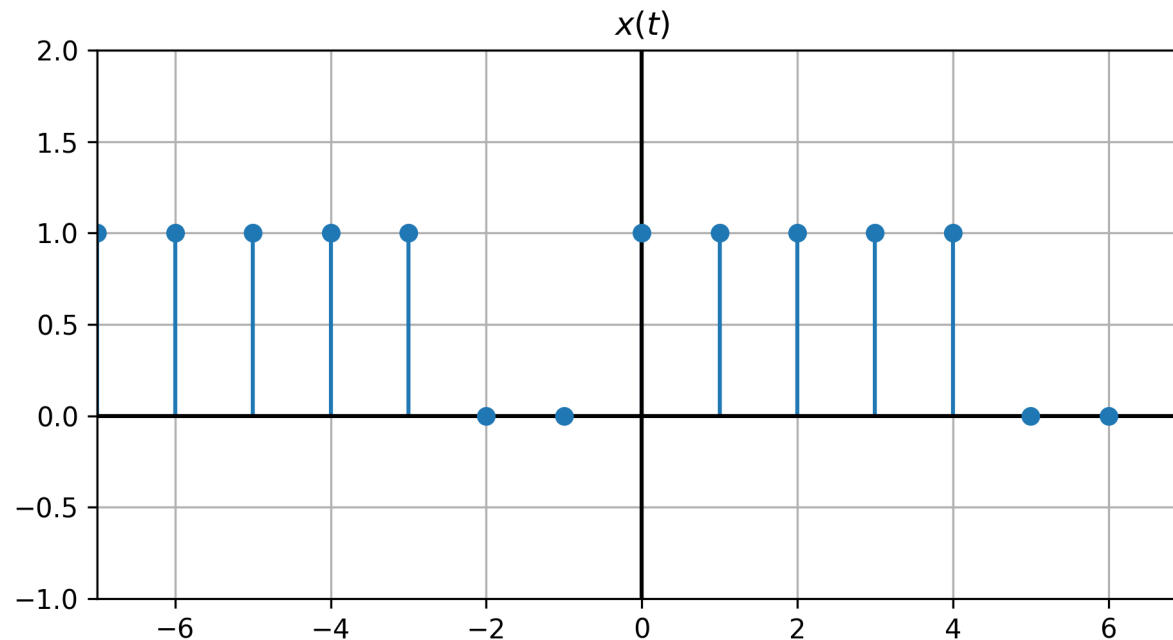


Coeficientes : $|k| \leq 0$



3.28. Determine los coeficientes de la serie de Fourier para cada una de las siguientes señales periódicas discretas. Grafique la magnitud y fase de cada conjunto de coeficientes a_k .

a) Cada $x[n]$ dibujada en la figura.



Solución: N=7

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n}$$

$$a_k = \frac{1}{7} \sum_{n=0}^4 \left(e^{-jk \frac{2\pi}{7}} \right)^n$$

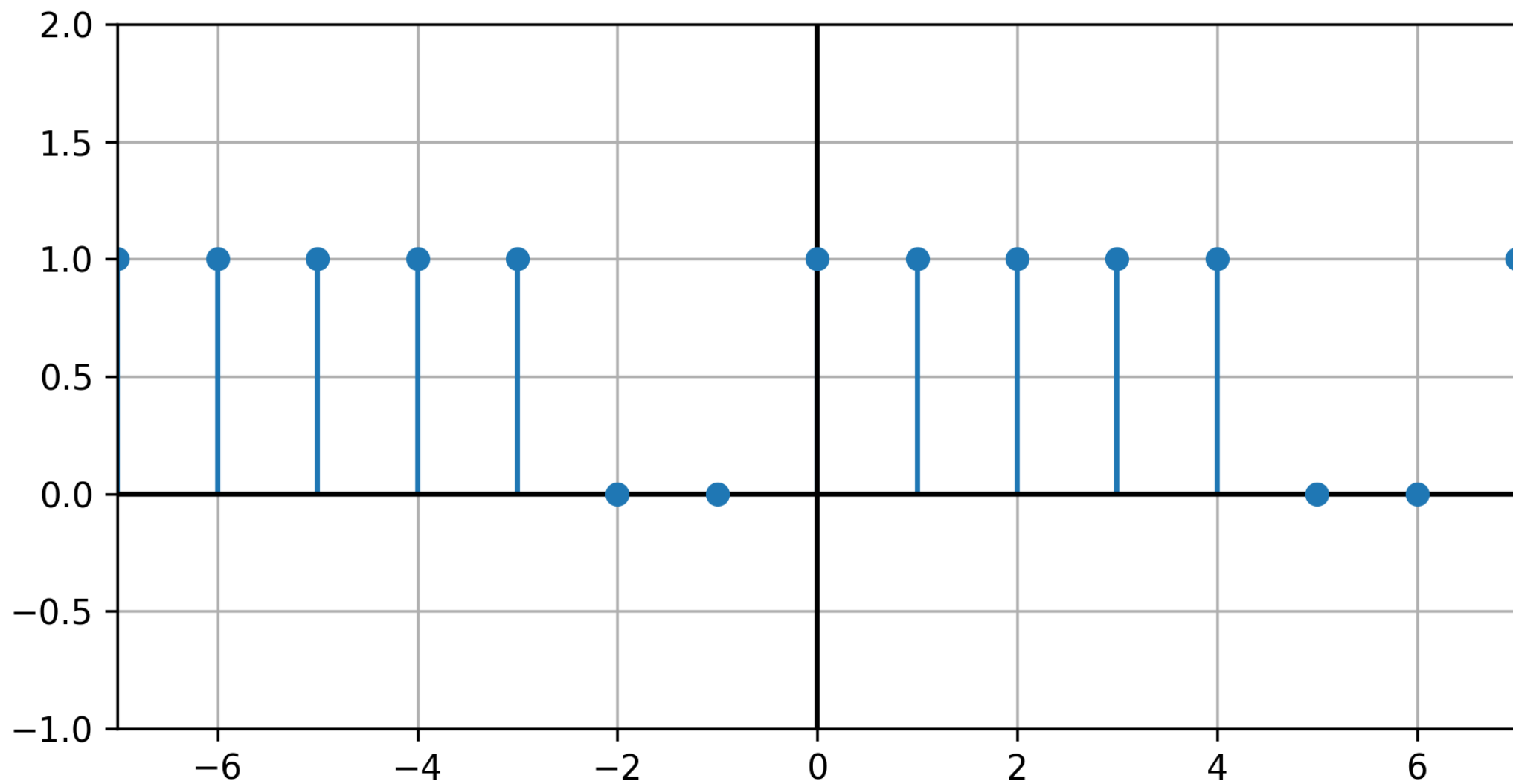
$$a_k = \frac{1}{7} \left[\frac{1 - \left(e^{-jk \frac{2\pi}{7}} \right)^5}{1 - e^{-jk \frac{2\pi}{7}}} \right]$$

Calculo Auxiliar

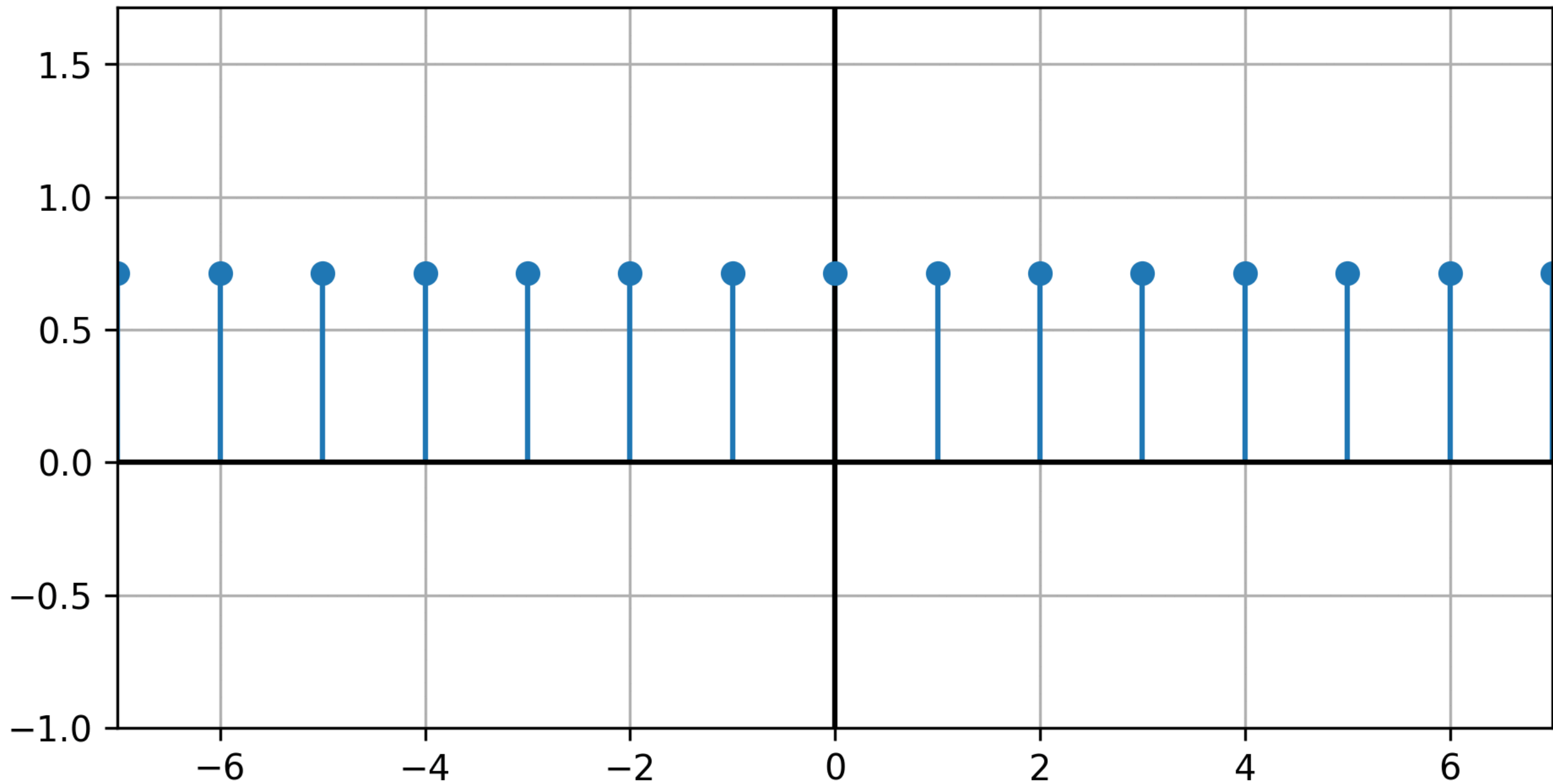
Serie geométrica:

$$\sum_{n=0}^b \alpha^n = \frac{1 - \alpha^{b+1}}{1 - \alpha}$$

$x[n]$



Coeficientes : $|k| \leq 0$



3.29. En cada uno de los siguientes incisos especificamos los coeficientes de la serie de Fourier de una señal que es periódica con periodo 8. Determine la señal $x[n]$ en cada caso.

$$\text{a) } a_k = \cos\left(\frac{k\pi}{4}\right) + \sin\left(\frac{3k\pi}{4}\right)$$

$$a) a_k = \cos\left(\frac{k\pi}{4}\right) + \sin\left(\frac{3k\pi}{4}\right)$$

Solución:

$$N=8$$

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\frac{2\pi}{N}n}$$

Calculo Auxiliar

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$x[n] = \sum_{k=0}^7 \left[\frac{e^{jk\frac{\pi}{4}} + e^{-jk\frac{\pi}{4}}}{2} + \frac{e^{j\frac{3k\pi}{4}} - e^{-j\frac{3k\pi}{4}}}{2j} \right] e^{kj\frac{2\pi}{8}n}$$

1 - Adicional) Encontrar la expresión en Serie de Fourier de la salida de un SLIT si:

$$x[n] = \text{sen} \left(\frac{\pi}{3} n \right) + \cos \left(3 \frac{\pi}{2} n + \frac{\pi}{4} \right)$$

$$h[n] = u[n - 1] - u[n - 7]$$

Solución:

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{j \frac{2\pi}{N} kn}$$

$$y[n] = \sum_{k=\langle N \rangle} b_k e^{j \frac{2\pi}{N} kn}$$

$$y[n] = \sum_{k=\langle N \rangle} a_k H\left(\frac{k}{N}\right) e^{j \frac{2\pi}{N} kn}$$



$$b_k = a_k H\left(\frac{k}{N}\right)$$

$$\Omega_0 = \frac{2\pi}{N}$$

$$H\left(\frac{k}{N}\right) = H\left(e^{jk \frac{2\pi}{N}}\right) = \sum_{n=-\infty}^{\infty} h[n] e^{-j \Omega_0 kn}$$

$$h[n] = u[n-1] - u[n-7] \quad \Rightarrow \quad H\left(\frac{k}{N}\right) = H\left(e^{jk\frac{2\pi}{N}}\right) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\Omega_0 kn}$$

$$H\left(\frac{k}{N}\right) = \sum_{n=1}^6 \left(e^{-j\Omega_0 k}\right)^n$$

$$H\left(\frac{k}{N}\right) = \frac{e^{-j\frac{2\pi}{N}k} - e^{-j\frac{2\pi}{N}k \cdot 7}}{1 - e^{-j\frac{2\pi}{N}k}}$$

Cálculo Auxiliar

$$\sum_{n=M}^N a^n = \frac{a^M - a^{N+1}}{1 - a}$$

$$\Omega_0 = \frac{2\pi}{N}$$

$$x[n] = \text{sen}\left(\frac{\pi}{3}n\right) + \cos\left(3\frac{\pi}{2}n + \frac{\pi}{4}\right) = \text{sen}\left(\frac{\overset{k=\pm 2}{2\pi}}{\underset{2.3.2}{2}}2n\right) + \cos\left(3\frac{\overset{k=\pm 9}{2\pi}}{\underset{2.2.3}{2}}3n + \frac{\pi}{4}\right)$$

Cálculo Auxiliar

$$\Rightarrow N_0 = 12$$

$$a_2 = \frac{1}{2j} \quad ; \quad a_{-2} = -\frac{1}{2j}$$

$$a_9 = \frac{e^{j\frac{\pi}{4}}}{2} \quad ; \quad a_{-9} = \frac{e^{-j\frac{\pi}{4}}}{2}$$

$$\text{sen}\left(\frac{2\pi}{N_0}kn\right) = \frac{e^{j\frac{2\pi}{N_0}kn} - e^{-j\frac{2\pi}{N_0}kn}}{2j}$$

$$\cos\left(\frac{2\pi}{N_0}kn + \frac{\pi}{4}\right) = \frac{e^{j\left(\frac{2\pi}{N_0}kn + \frac{\pi}{4}\right)} + e^{-j\left(\frac{2\pi}{N_0}kn + \frac{\pi}{4}\right)}}{2}$$

$$H\left(\frac{k}{N_0}\right) = \frac{e^{-j\frac{2\pi}{N_0}k} - e^{-j\frac{2\pi}{N_0}k.7}}{1 - e^{-j\frac{2\pi}{N_0}k}} \Rightarrow H\left(\frac{\pm 2}{12}\right) = 0$$

$$H\left(\frac{9}{12}\right) = -1 + j$$

$$H\left(\frac{-9}{12}\right) = -1 - j$$

$$y[n] = \sum_{k=\langle N \rangle} a_k H\left(\frac{k}{N}\right) e^{j\frac{2\pi}{N}kn}$$

$$y[n] = a_9 H\left(\frac{9}{12}\right) e^{j\frac{9.2\pi}{12}n} + a_{-9} H\left(\frac{-9}{12}\right) e^{-j\frac{9.2\pi}{12}n}$$

$$y[n] = \frac{e^{j\frac{\pi}{4}}}{2} (-1 + i) e^{j\frac{9.2\pi}{12}n} + \frac{e^{-j\frac{\pi}{4}}}{2} (-1 - i) e^{-j\frac{9.2\pi}{12}n}$$

$$y[n] = \frac{e^{j\frac{\pi}{4}}}{2} (\sqrt{2} e^{j\frac{3\pi}{4}}) e^{j\frac{9.2\pi}{12}n} + \frac{e^{-j\frac{\pi}{4}}}{2} (\sqrt{2} e^{j\frac{3\pi}{4}}) e^{-j\frac{9.2\pi}{12}n}$$

$$y[n] = \frac{1}{\sqrt{2}} (e^{j\pi}) e^{j\frac{9.2\pi}{12}n} + \frac{1}{\sqrt{2}} (e^{j\frac{\pi}{2}}) e^{-j\frac{9.2\pi}{12}n}$$

$$y[n] = \frac{1}{\sqrt{2}} \left(-e^{j\frac{9.2\pi}{12}n} + j e^{-j\frac{9.2\pi}{12}n} \right)$$

Cálculo Auxiliar

$$-1 + i = \sqrt{2} e^{j\frac{3\pi}{4}}$$

$$-1 - i = \sqrt{2} e^{-j\frac{3\pi}{4}}$$

$$e^{j\pi} = -1$$

$$e^{j\frac{\pi}{2}} = j$$